

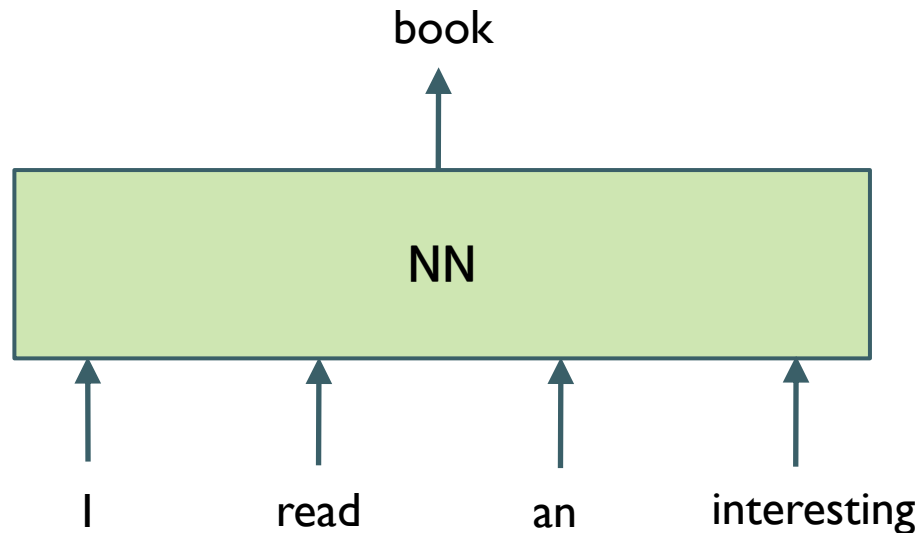
NLP + Transformer

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이 지 형

Next Word Prediction

▶ Example

I read an interesting → book



Next Word Prediction

▶ Training Data

Crawled Text

I read an interesting book. It was about the life of a famous scientist who made groundbreaking discoveries in the field of biology. The book provided a detailed account of his experiments, challenges, and triumphs. I was particularly fascinated by how his persistence and dedication led to significant advancements in science. It was an inspiring read, and I learned a lot about the scientific process and the importance of never giving up.

Training Data

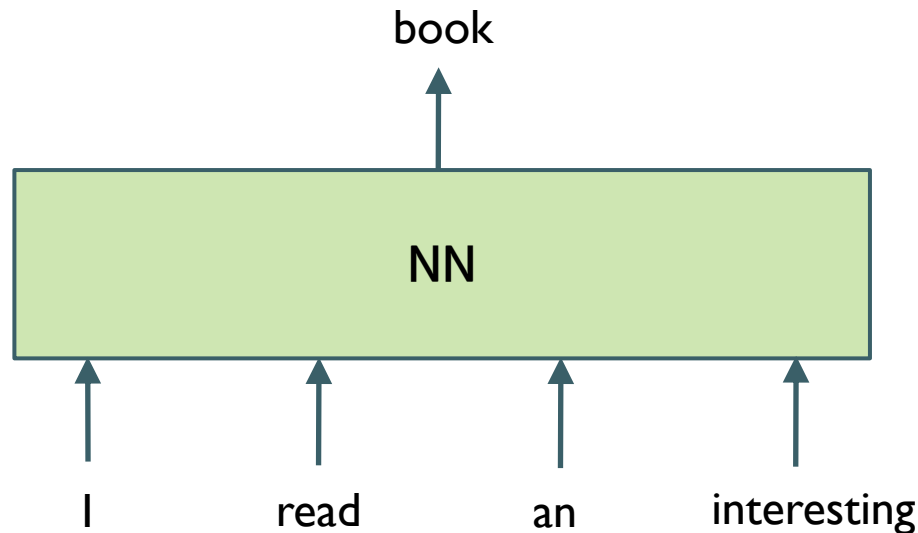
(NA NA NA I → read)
(NA NA I read → an)
(NA I read an → interesting)
(I read an interesting → book)
(read an interesting book →.)
(an interesting book. → It)
(interesting book. It → was)
(book. It was → about)

...

Next Word Prediction

▶ How to Handle Texts

I read an interesting → book



Next Word Prediction

► How to Handle Texts

(A bird is → flying)
(A dog is → running)
(A fish is → swimming)
... ..

Unique words

a an bird dog fish is flying running swims ...

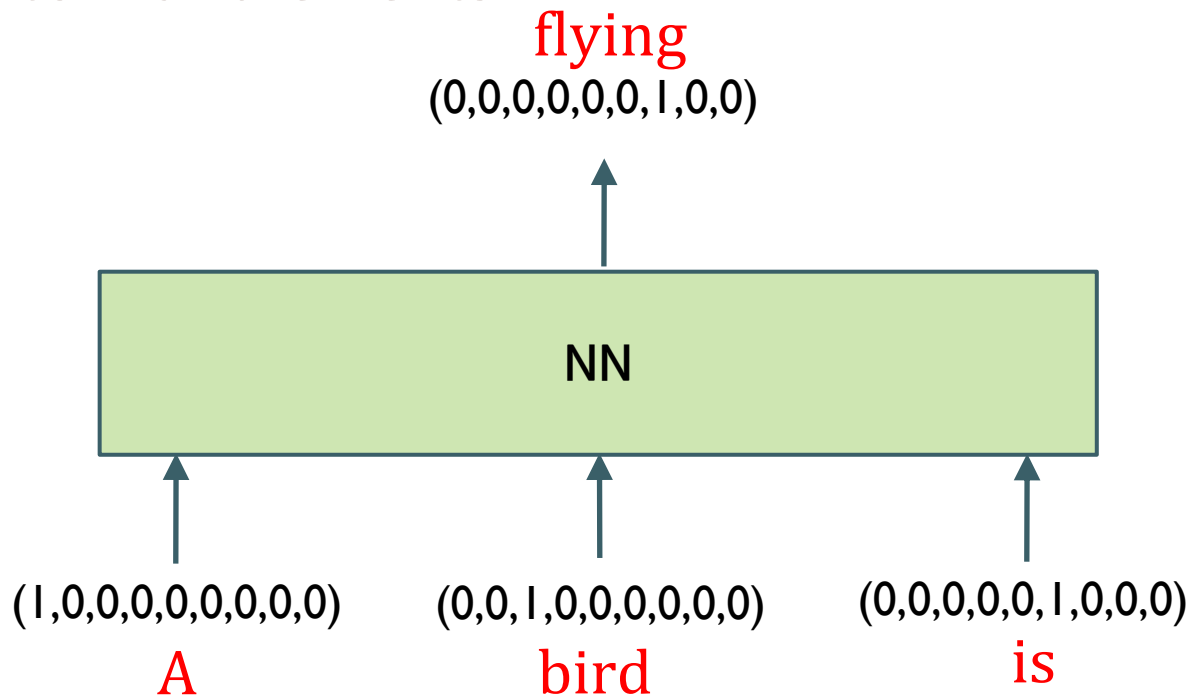
A (1 , 0 , 0 , 0 , 0 , 0 , 0 , 0 , 0 , 0 , ...)

bird (0 , 0 , 1 , 0 , 0 , 0 , 0 , 0 , 0 , 0 , ...)

is (0 , 0 , 0 , 0 , 0 , 1 , 0 , 0 , 0 , 0 , ...)

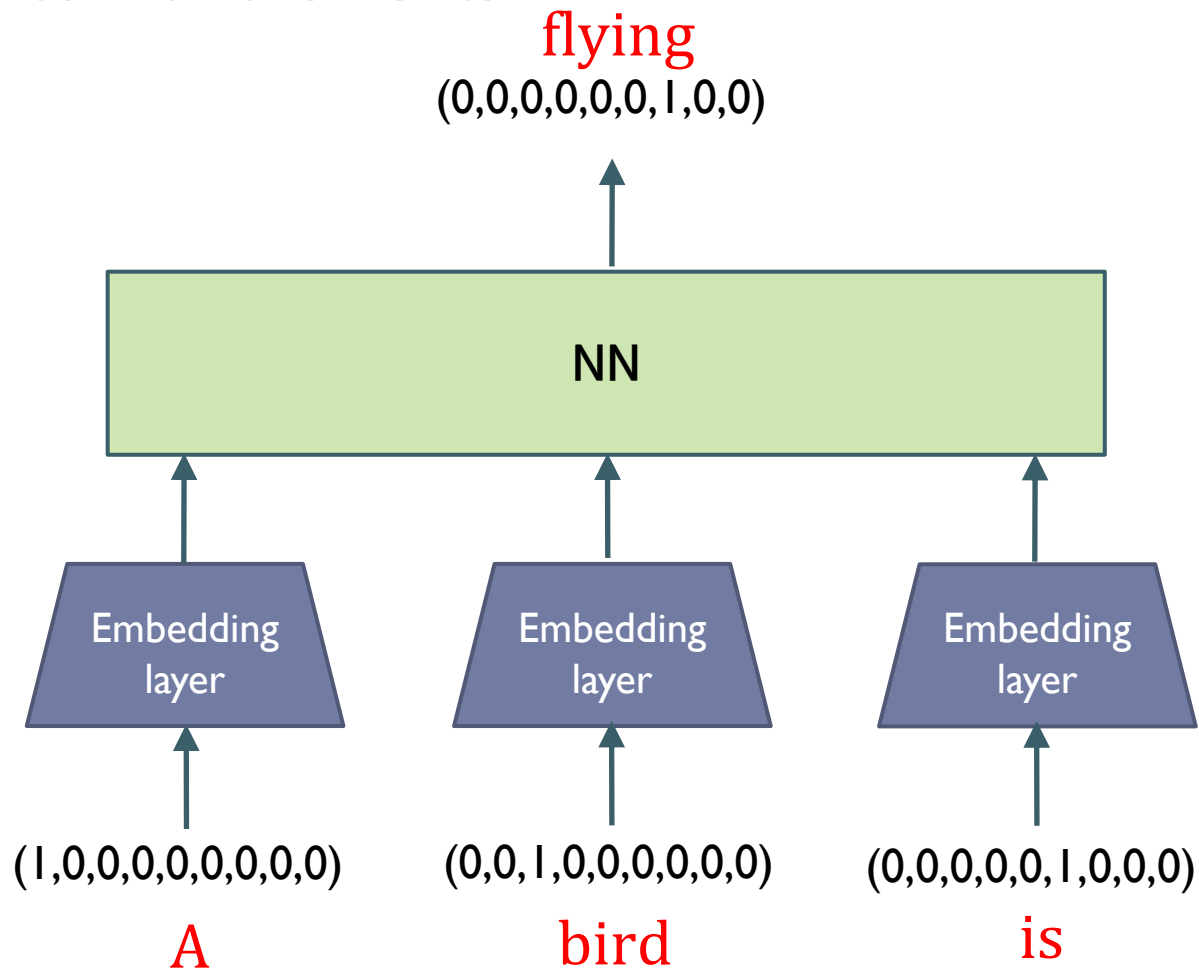
Next Word Prediction

▶ How to Handle Texts



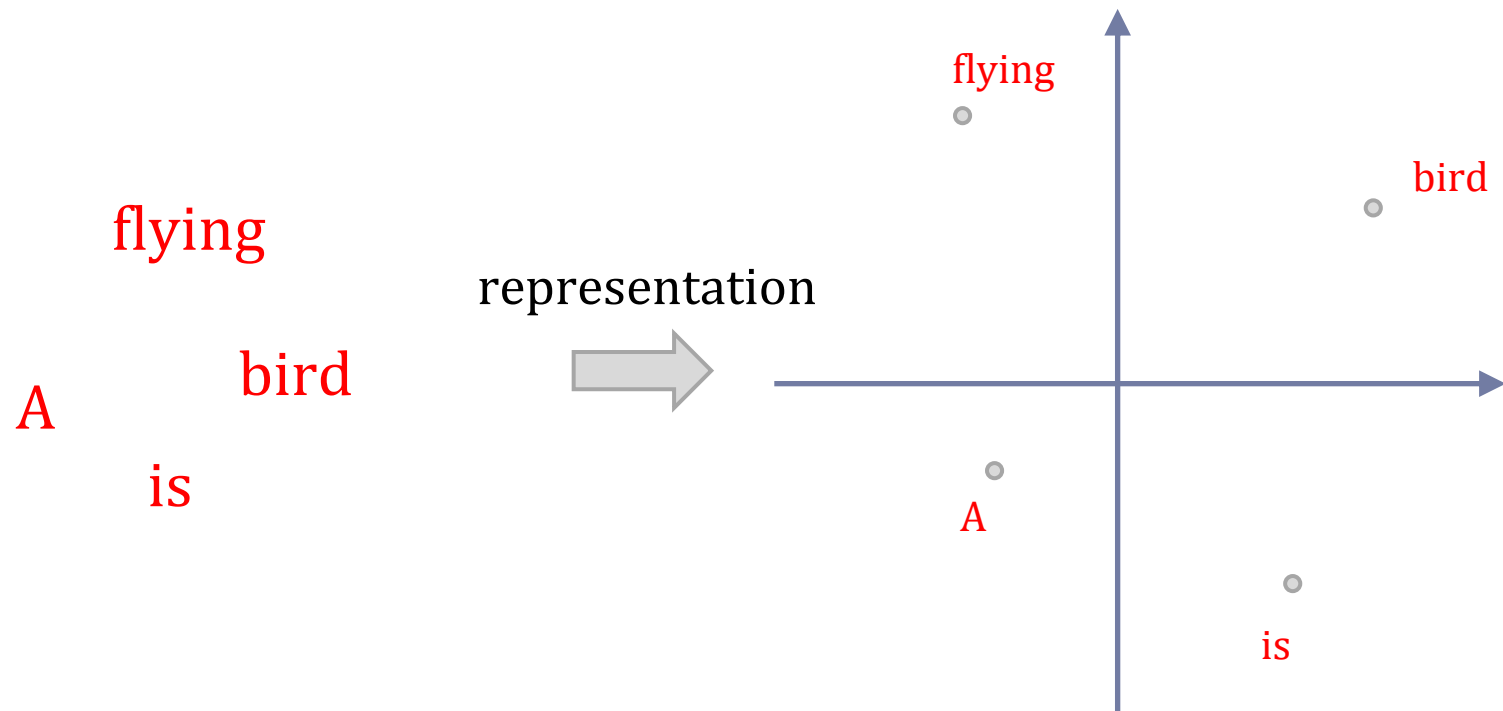
Next Word Prediction

► How to Handle Texts



Next Word Prediction

► Representation, Embedding



Values in non-Euclidean space

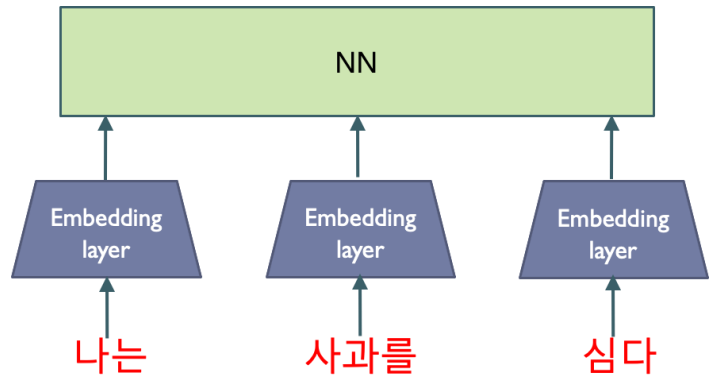
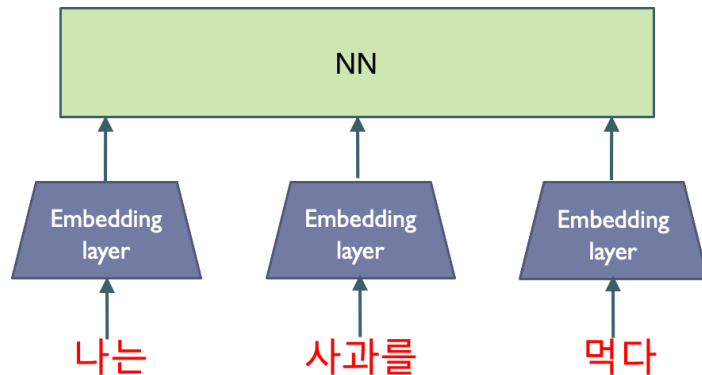
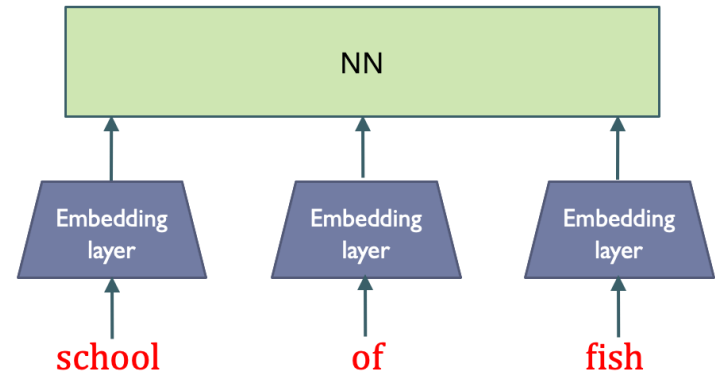
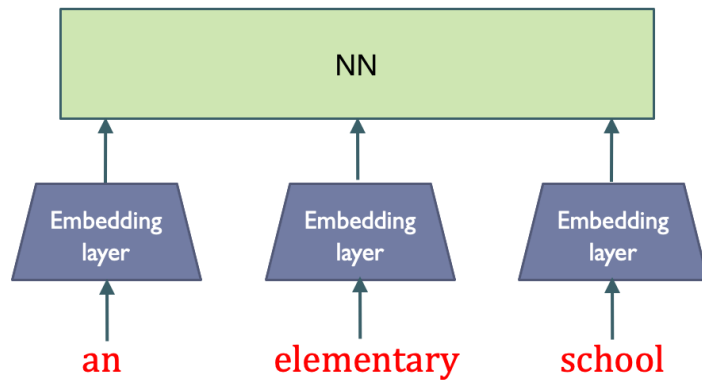
Values in Euclidean space



Attention Mechanism

Word Representation

► Enough? Embedding Layer

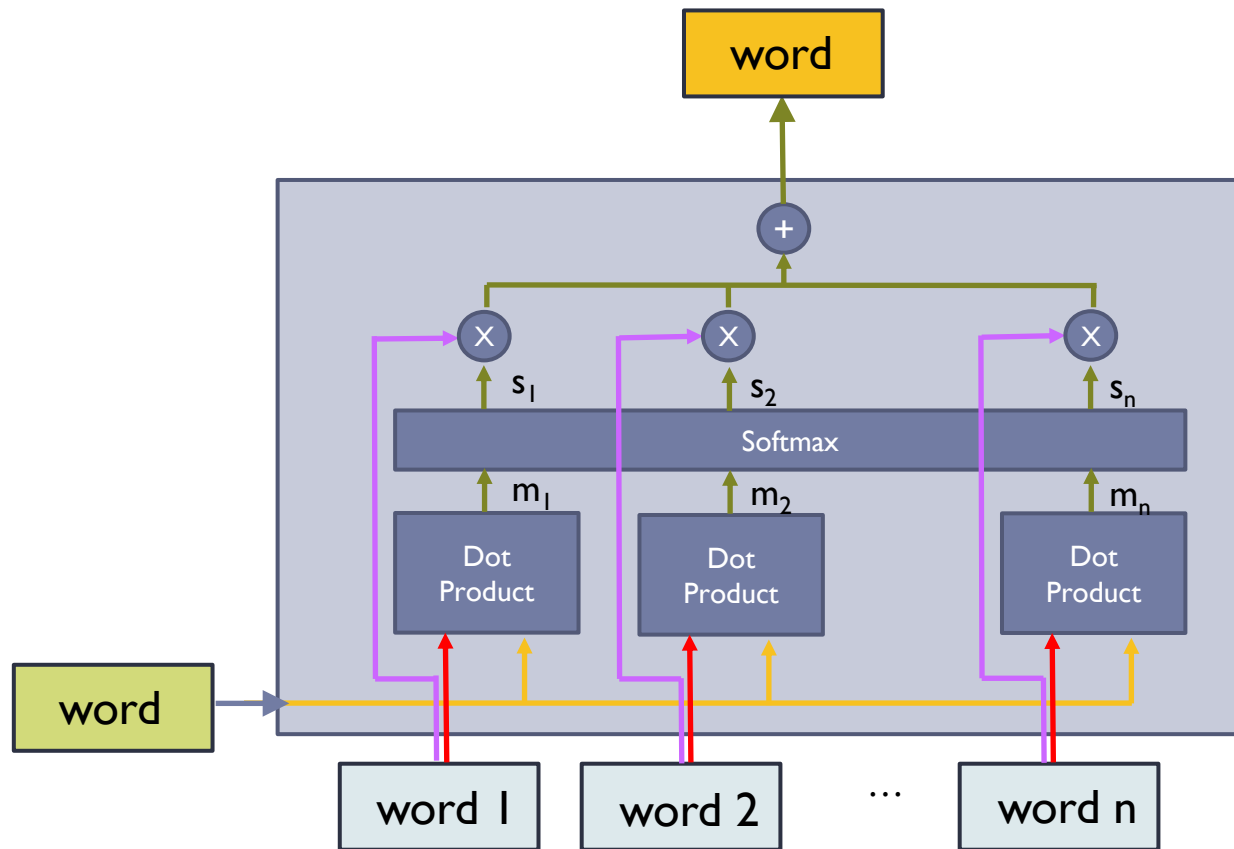


Word Representation

- ▶ Words have their own meanings independently
 - ▶ But in a sentence, they often take on different, more specific meanings depending on context.
 - ▶ Let's represent a word by blending it with other words in the sentence
- This leads to the concept of Attention

Attention Mechanism

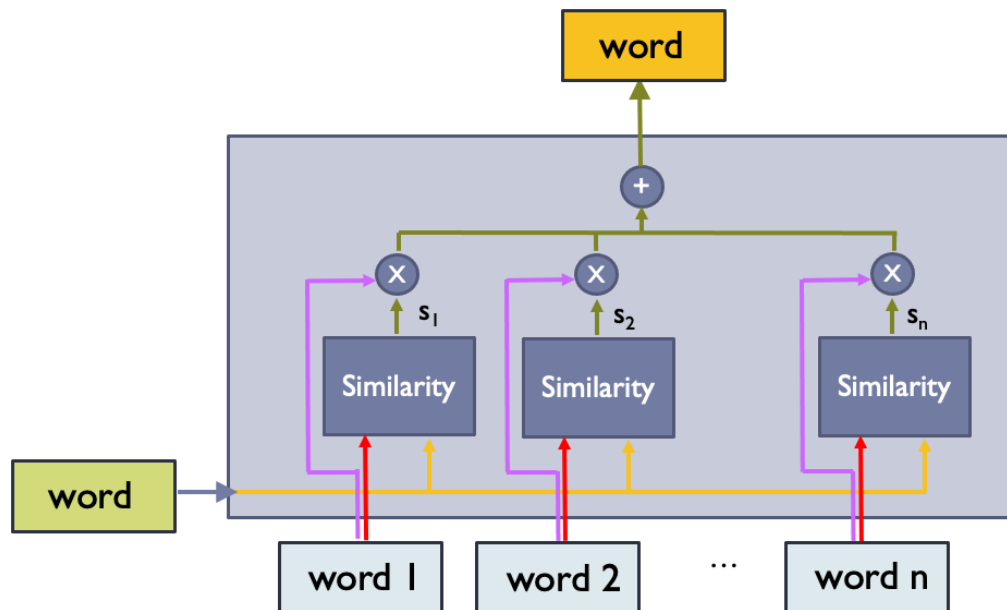
► Overview



Context Consideration

▶ Generating Word Representation with Context

$$\boxed{\text{word}} = s_1 \times \boxed{\text{word 1}} + s_2 \times \boxed{\text{word 2}} + \cdots + s_n \times \boxed{\text{word n}}$$

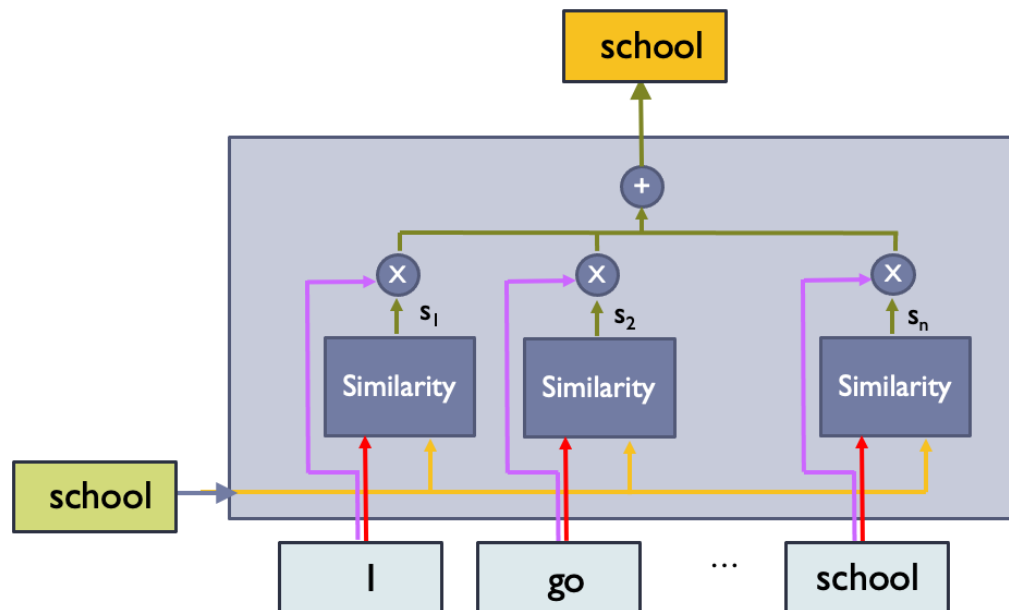


Context Consideration

▶ Generating Word Representation with Context

▶ Example

$$\boxed{\text{school}} = s_1 \times \boxed{\text{I}} + s_2 \times \boxed{\text{go}} + \dots + s_n \times \boxed{\text{school}}$$

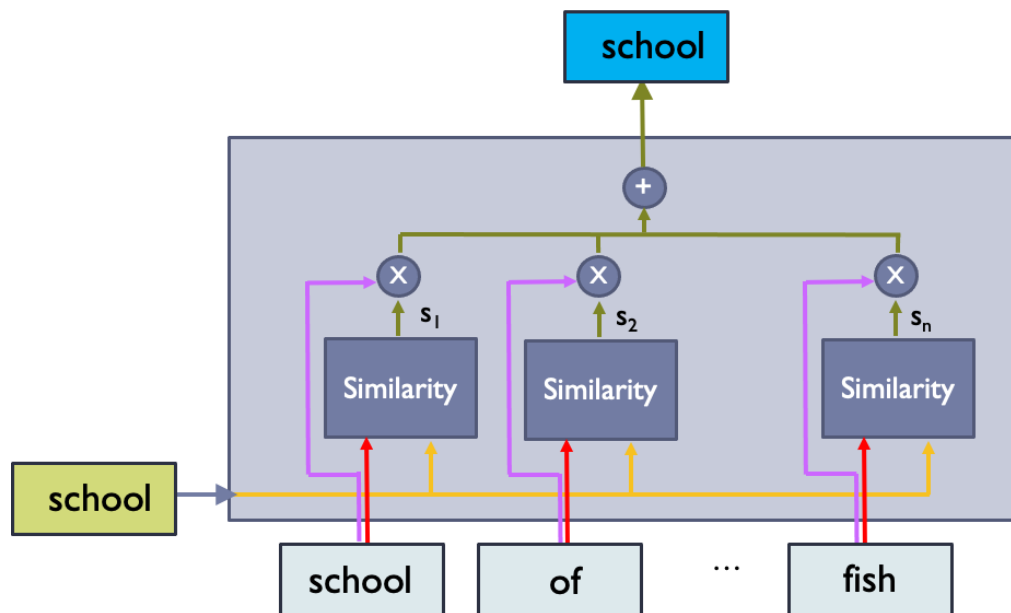


Context Consideration

▶ Generating Word Representation with Context

▶ Example

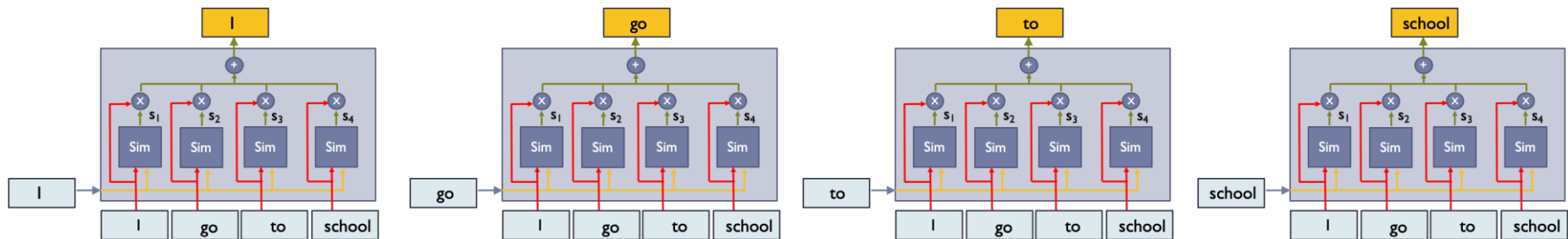
$$\boxed{\text{school}} = s_1 \times \boxed{\text{school}} + s_2 \times \boxed{\text{of}} + \cdots + s_n \times \boxed{\text{fish}}$$



Self-Attention

- ▶ Context: The words in the same sentence

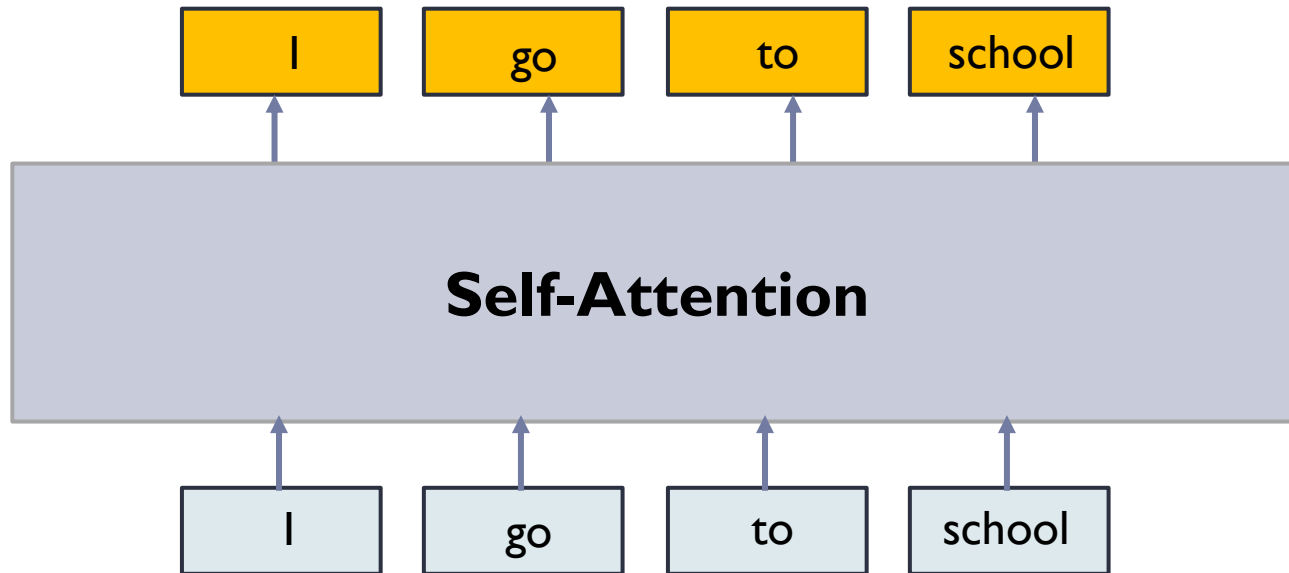
I go to school



I go to school

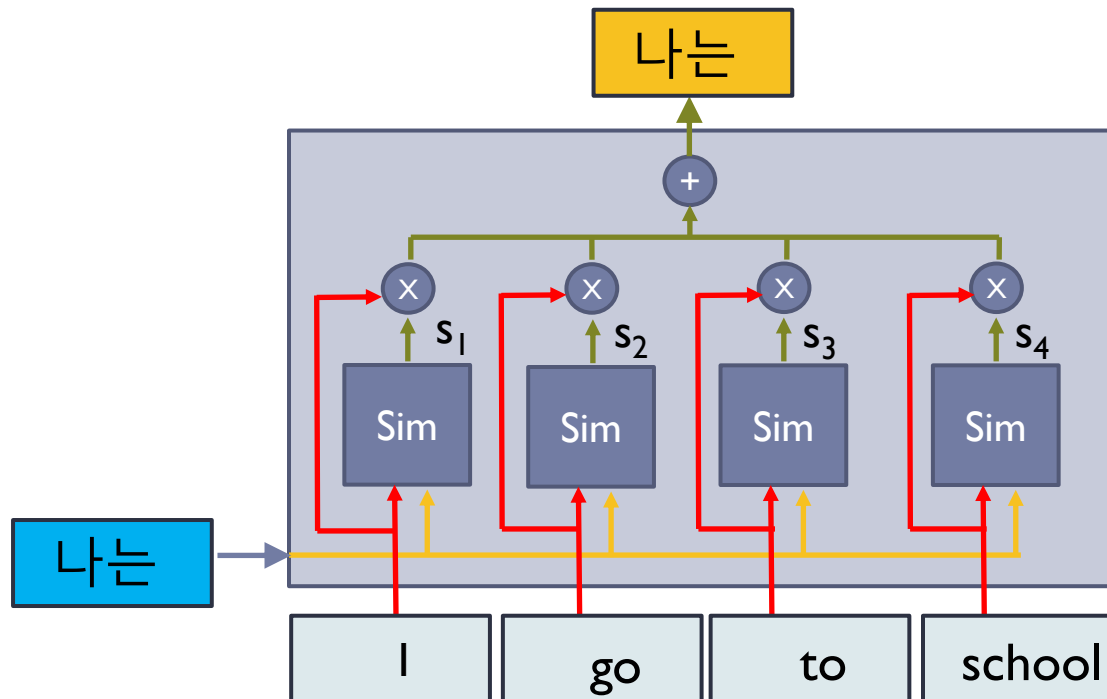
Self-Attention

- ▶ **Context:** The words in the same sentence



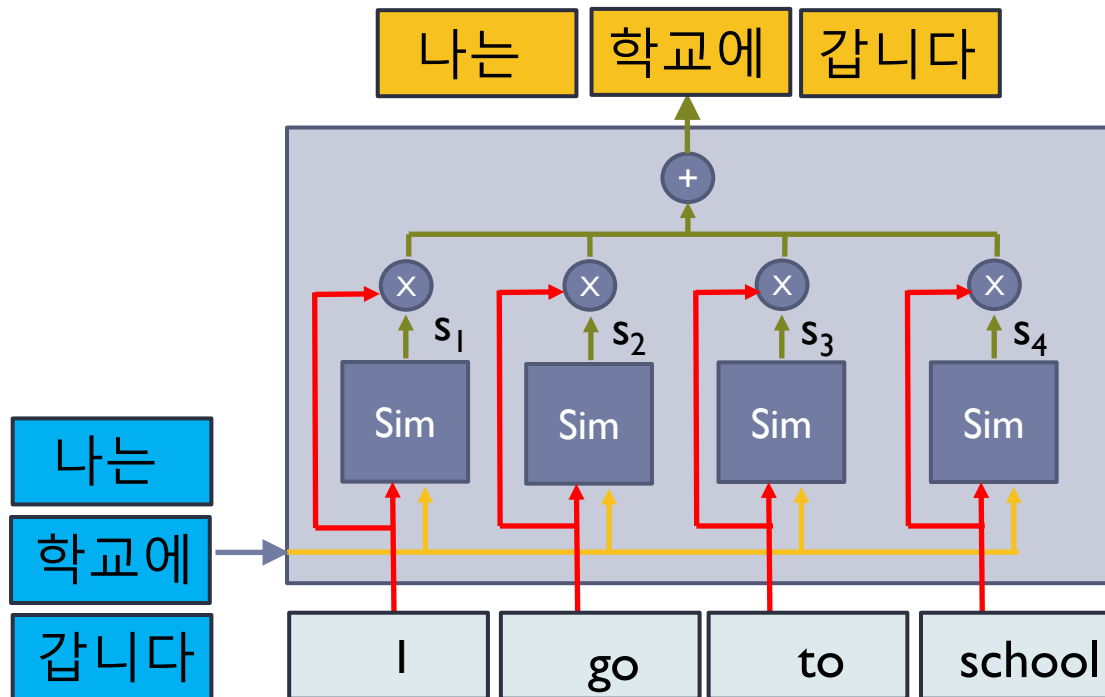
Cross-Attention

- ▶ **Context:** The words in the other sentence

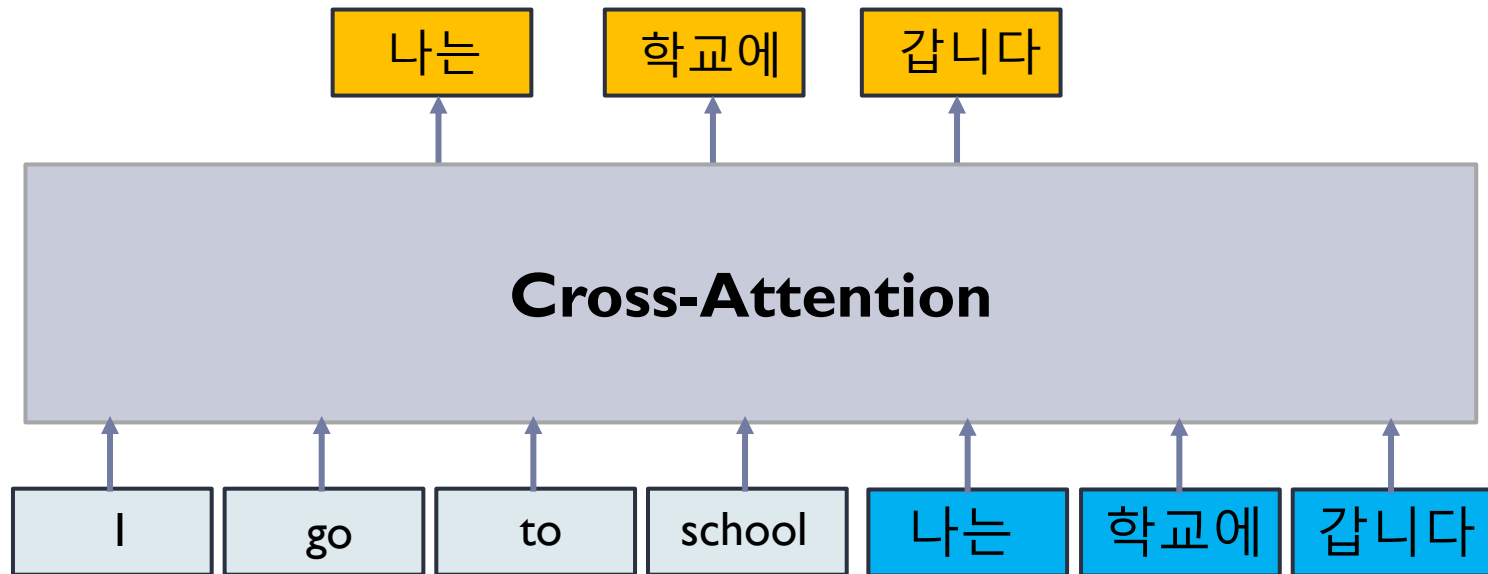


Cross-Attention

- ▶ **Context:** The words in the other sentence



Cross-Attention





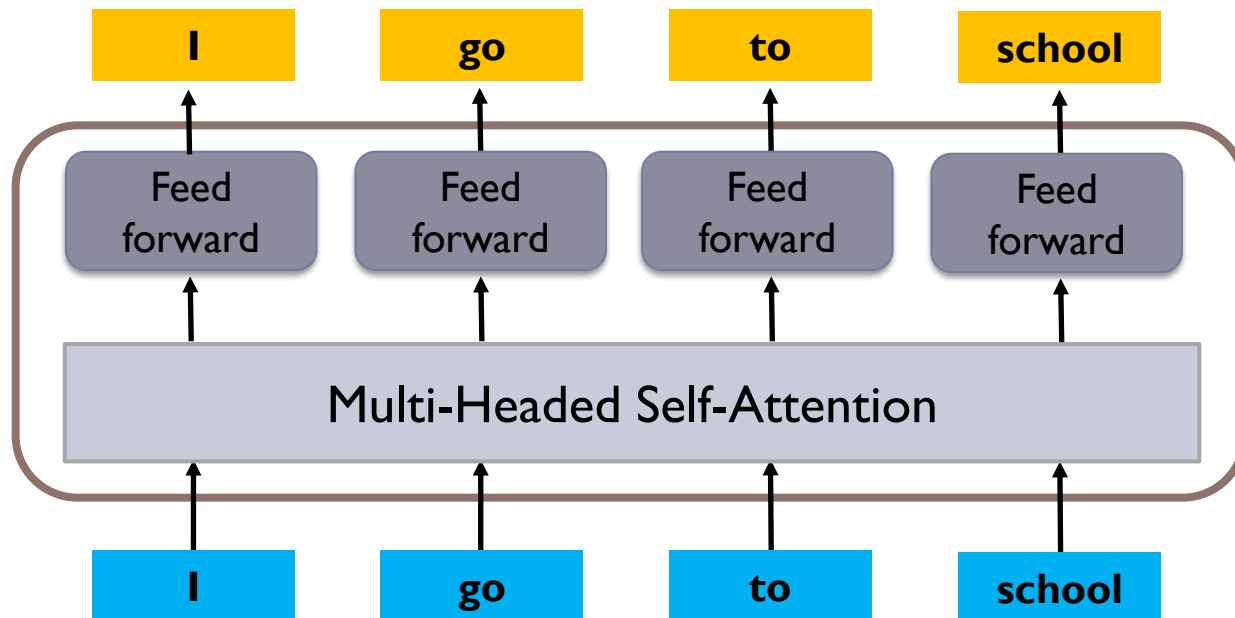
Trasformer

Transformer Encoder

▶ Structure

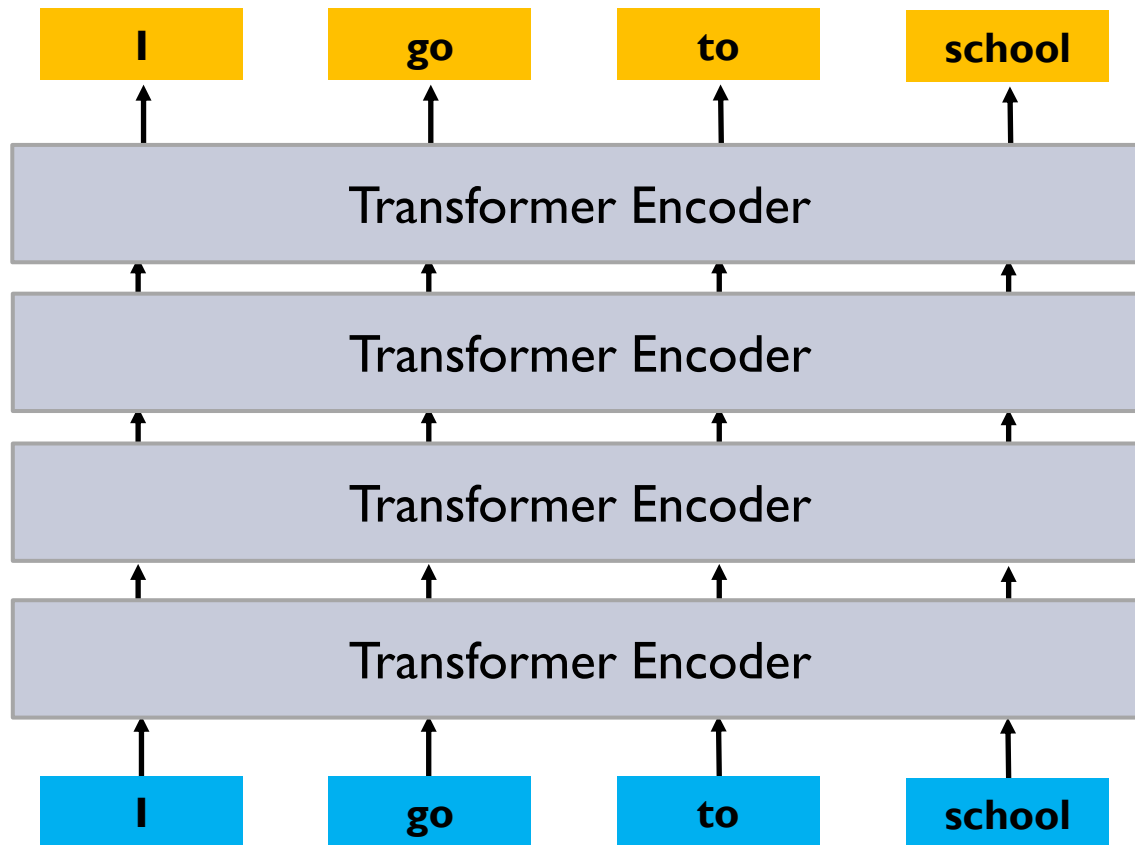
- ▶ Self-Attention + Multi-Head + Position Encoding + Feed forward NN layer

▶ More Sophisticated Semantic Processing



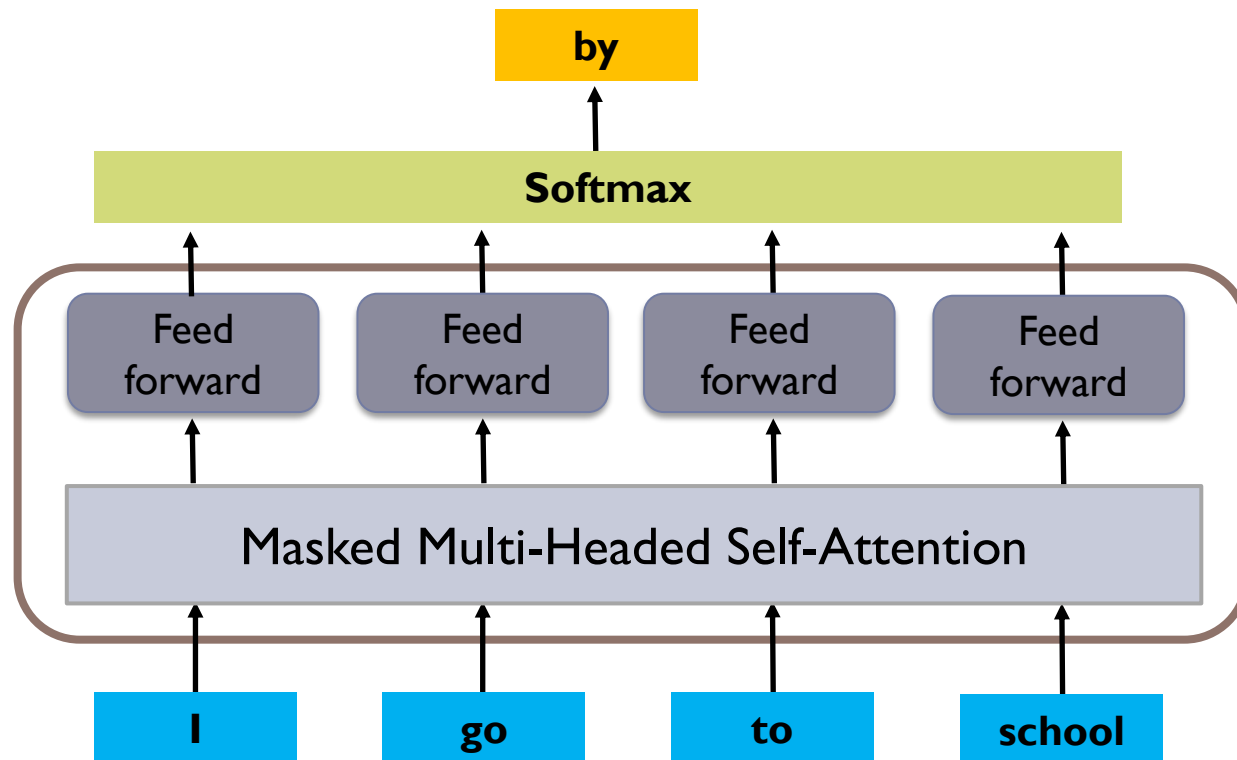
Transformer Encoder

- ▶ Word Representation Reflecting Sentence-Level Semantic



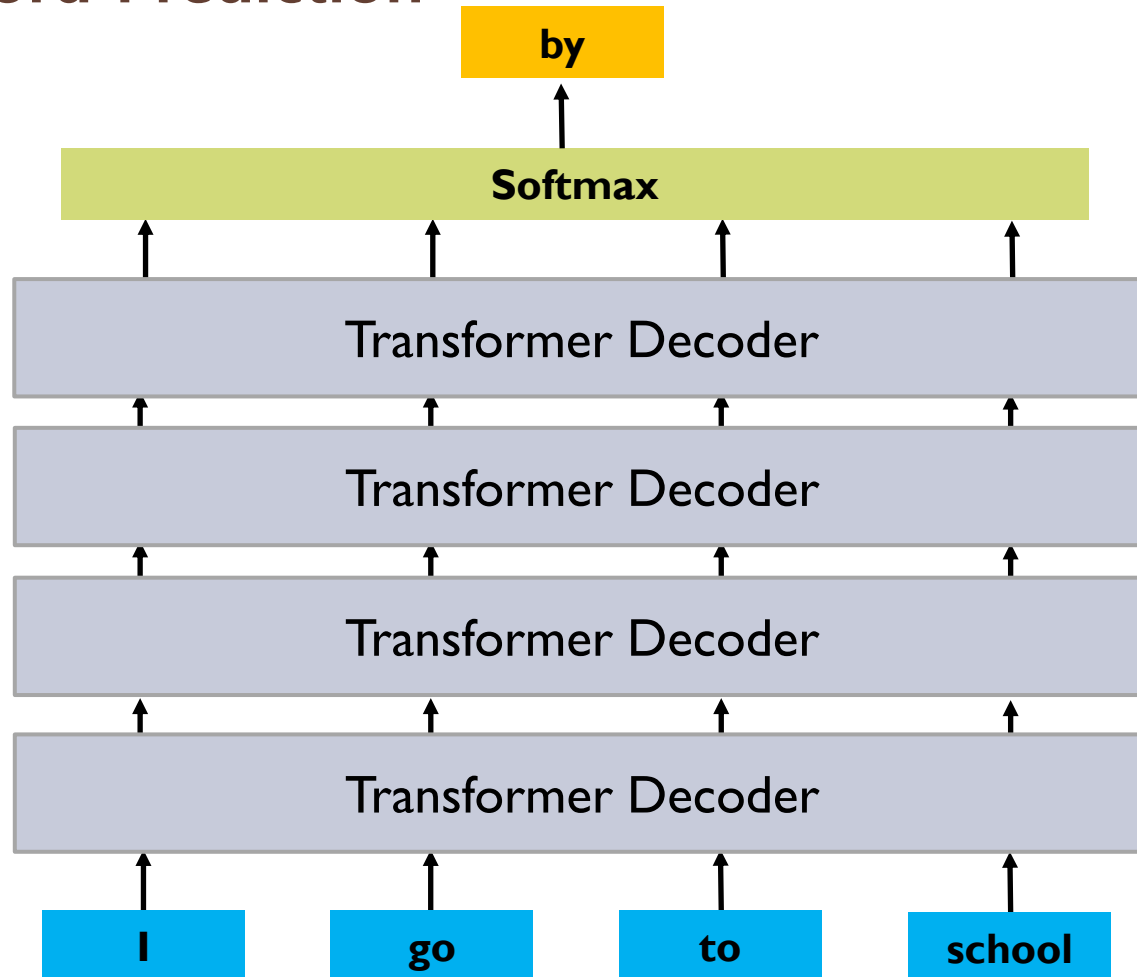
Transformer Decoder

- ▶ Structure: Similar to Encoder
- ▶ Prediction of next word



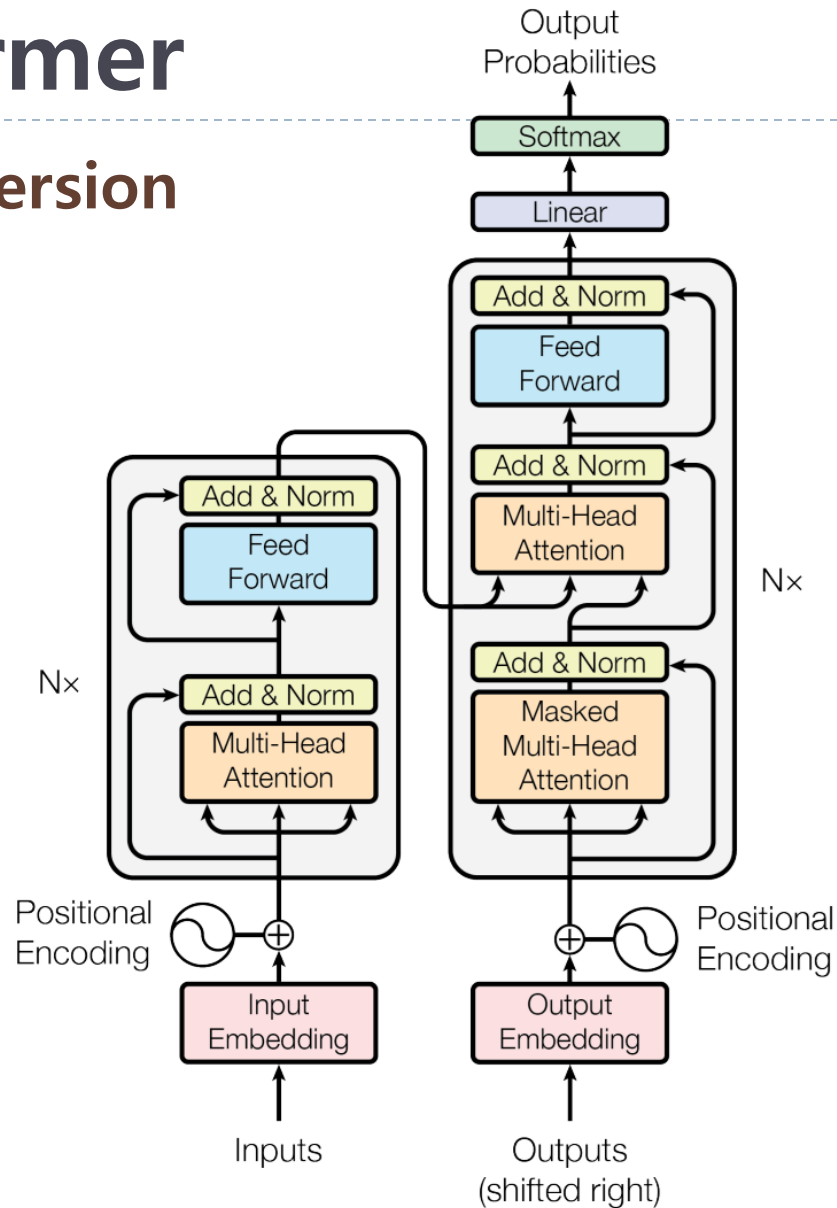
Transformer Decoder

▶ Next Word Prediction



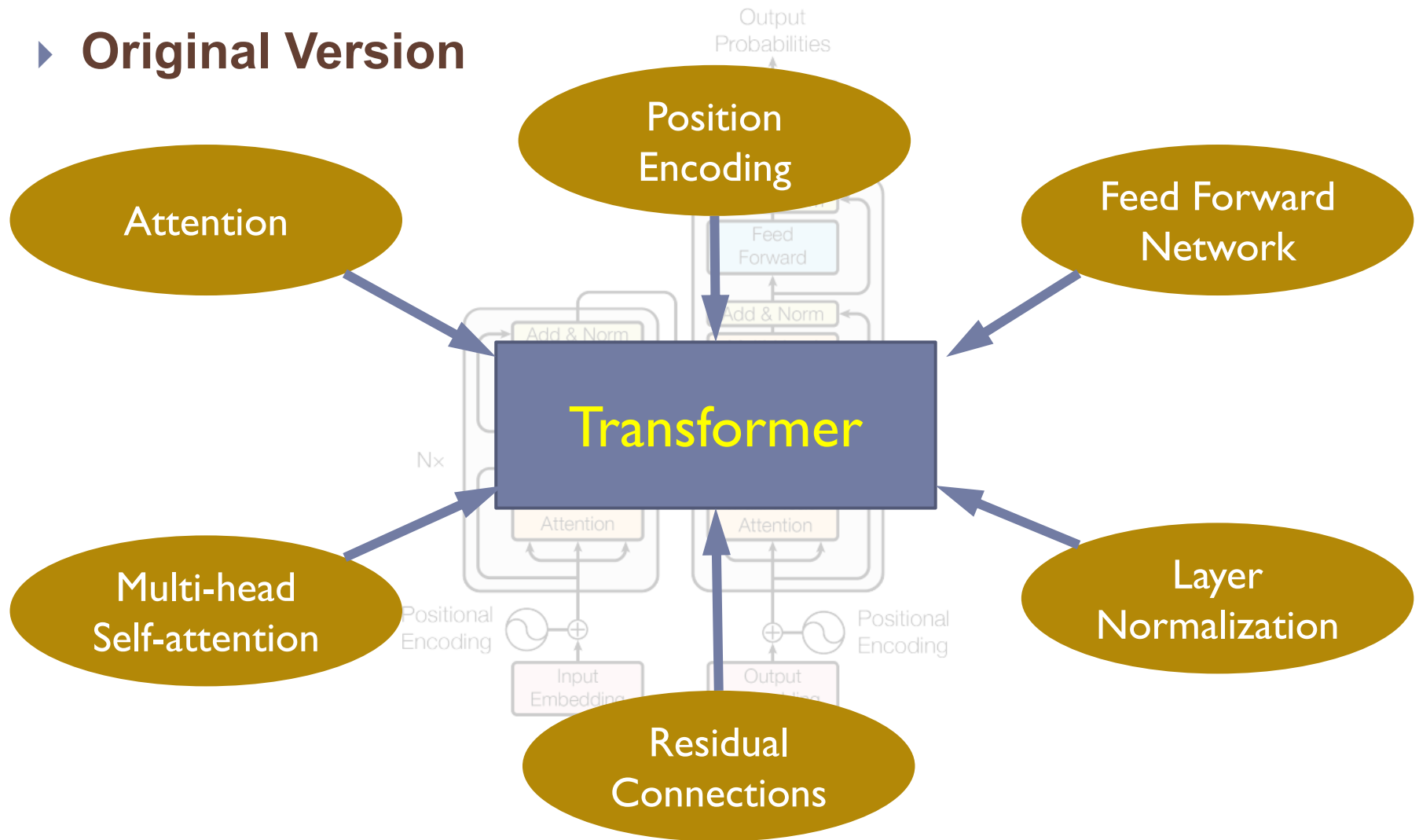
Transformer

► Original Version



Transformer

► Original Version





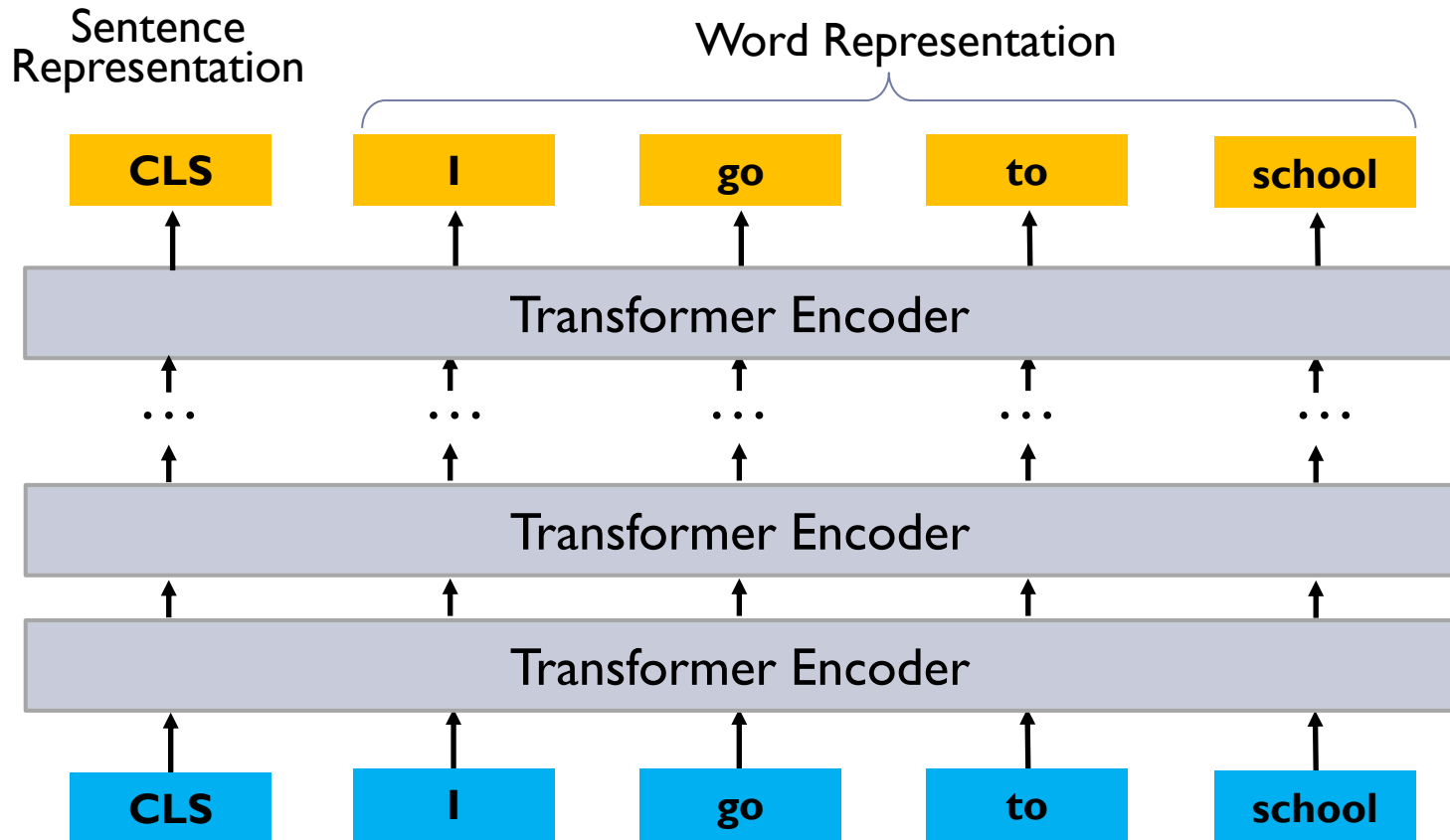
BERT, and GPT

BERT Model

- ▶ **BERT (Bidirectional Encoder Representations from Transformers)**
- ▶ **A powerful language understanding model**
- ▶ **Developed by Google in 2018**
- ▶ **Generating**
 - ▶ Word Representations Highly Processed considering Context
 - ▶ Sentence Representation
- ▶ **Pretrained Models**
 - ▶ KoBERT (by SKT): 한국어 위키백과 + 뉴스 기사로 학습됨
 - ▶ KorBERT (by ETRI)

BERT Model

► Structure: CLS token + Transformer Encoders



BERT Model

▶ Usage of CLS token

▶ Text classification

- ▶ Spam detection
- ▶ News categorization

▶ Sentiment analysis

- ▶ "I absolutely loved this product!" → Positive
- ▶ "The service was terrible and slow." → Negative

▶ Semantic similarity

- ▶ Two sentences have similar meaning or not

BERT Model

▶ Usage of Word Representation

▶ Named Entity Recognition (NER): 고유명사 찾기

▶ "Apple released a new product." → O, X, X, X, X

▶ Part-of-Speech Tagging: 품사 찾기

▶ "Apple released a new product" → noun, verb, article, adjective, noun

▶ Semantic Role Labeling: 문법적 역할 찾기

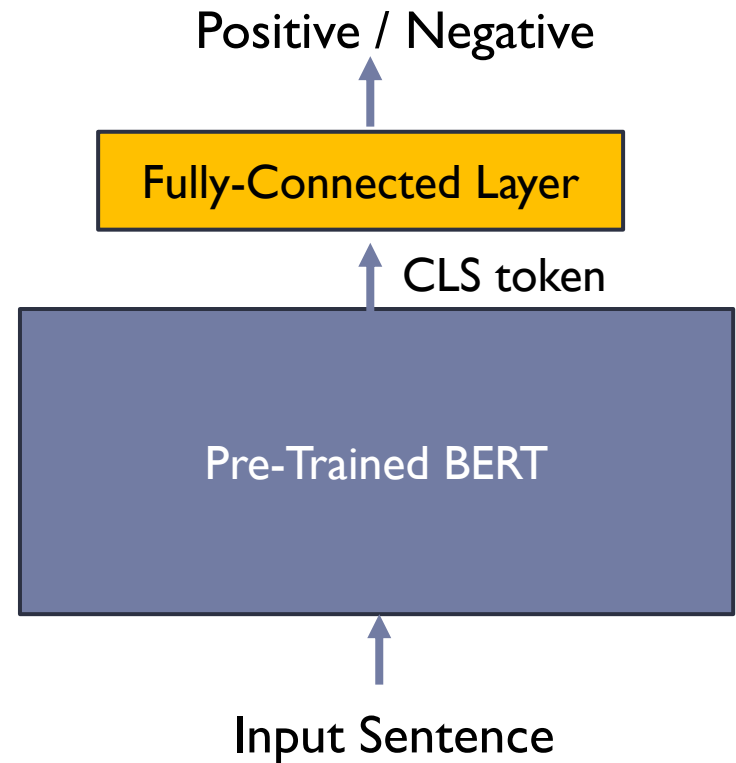
▶ "Apple released a new product" → subject, action, object

BERT Model

▶ Example: Sentiment Analysis

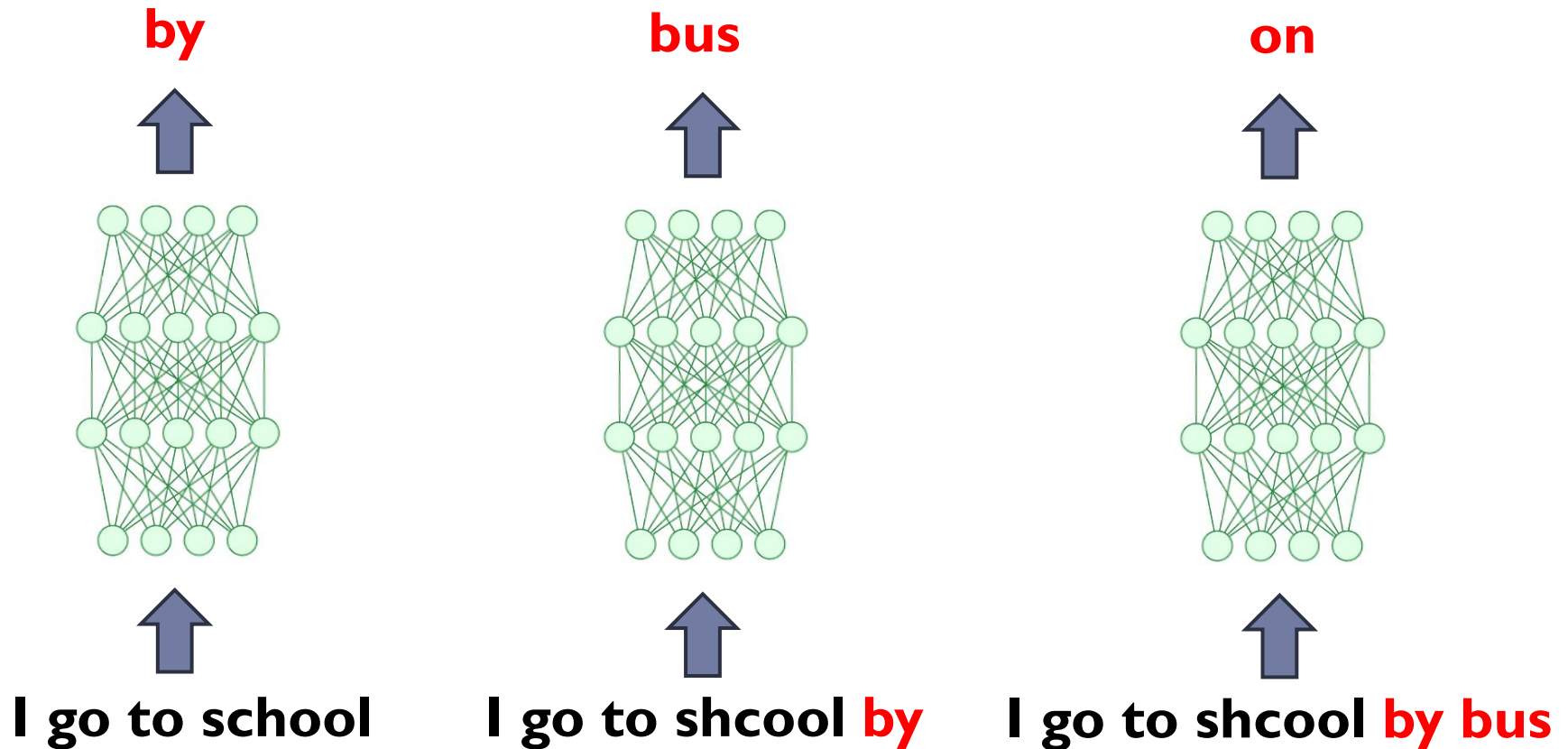
- ▶ CLS token에 NN layer를 붙인 후에
- ▶ 데이터를 이용하여 Fine-Tuning

이 제품 정말 좋아요. →	긍정
배송이 너무 느려요 →	부정
생각보다 괜찮았어요 →	긍정
기대 이하입니다 →	부정
가성비 최고 →	긍정
디자인 진짜 별로 →	부정



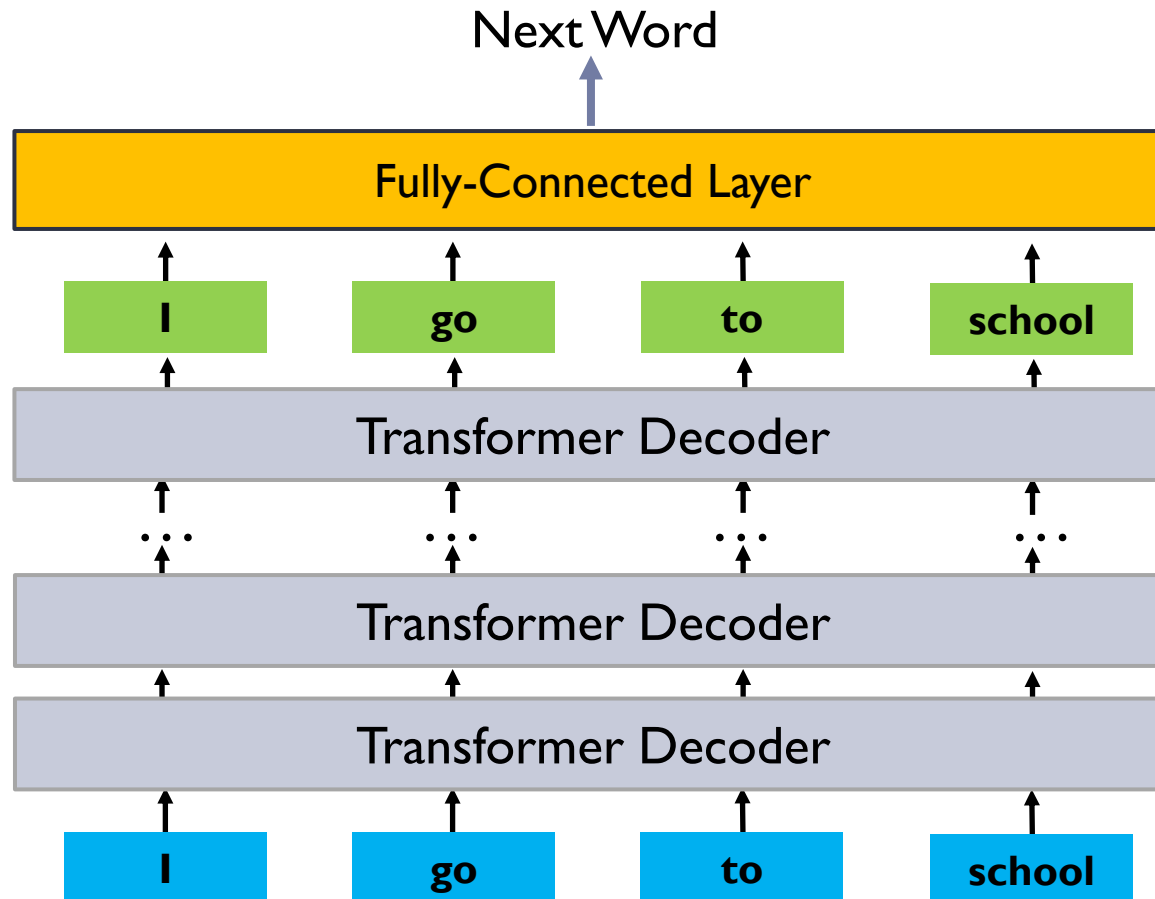
GPT Model

▶ Word Generation



GPT Model

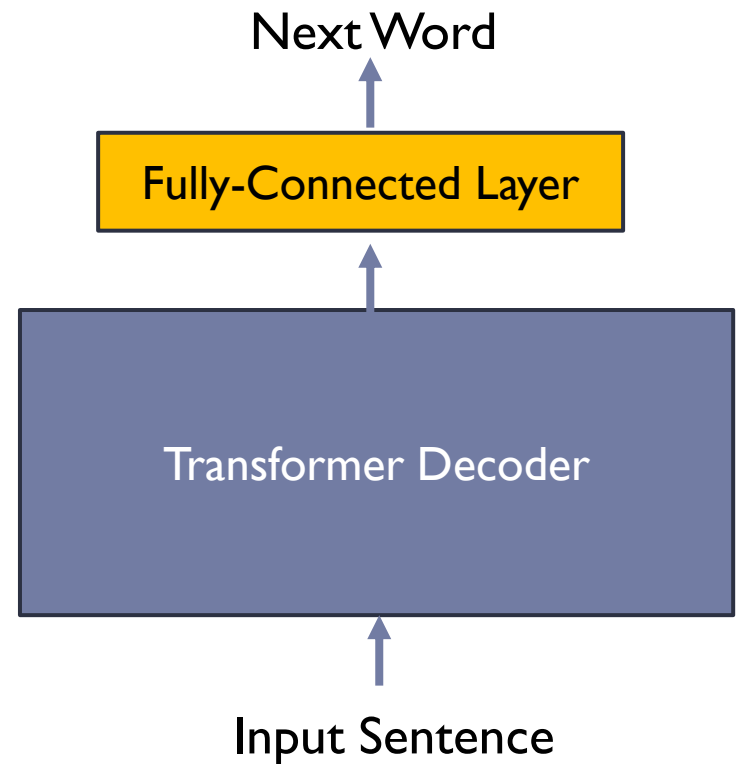
▶ Transformer Encoder + Fully-Connected Layer



GPT Model

▶ Example: English to Korean Translation

I go to school → 나는
I to to school 나는 → 학교에
I to to school 나는 학교에 → 갑니다
A bird is flying → 새가
A bird is flying 새가 → 날고
A bird is flying 새가 날고 → 있어요



Question and Answer